Exercise Sheet 0: Introduction to PyTorch

Due on 24.04.2026, 10:00

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Important Notes:

1. **Email**: Check your Moodle-registered email regularly. All course notifications will be sent via Moodle.
2. **Due Date**: Exercises are usually uploaded one week before the deadline (see top of the sheet).
3. **Submission**: Submit a ZIP file containing your code and a single PDF report. Both files must include your surname and matriculation number (e.g. *Surname-MatriculationNumber.zip*). You may export your report from Jupyter[1], but a separate `.py` file is still required. **Incorrectly named submissions will not be graded.**

General Information:
Exercises must be completed in Python 3 (version 3.8+). You may use standard libraries or Anaconda. We will use PyTorch[2] as the main deep learning framework. For questions, use the student forum on Moodle. For technical issues (e.g., upload problems), contact the person listed at the top of the sheet. These exercises assume basic Python knowledge. If needed, review a tutorial first. The following libraries will be useful:

- **SciPy**[3] – scientific computing
- **NumPy**[4] – array operations
- **Matplotlib**[5] – plotting
- **scikit-learn**[6] – basic ML algorithms
- **Pillow**[7] – image processing (note: only accepts `uint8` arrays with values in [0, 255])

[1] https://jupyter.org/
[2] https://pytorch.org/
[3] https://www.scipy.org/
[4] https://numpy.org/
[5] https://matplotlib.org/
[6] https://scikit-learn.org/stable/
[7] https://pillow.readthedocs.io/en/stable/

Task 1: Introduction to PyTorch (8P)

Background:
In this task, you will get familiar with basic PyTorch tensor operations. Use the notebook `exercise0.ipynb` and fill all `# TODO` cells. Keep your code short and clear.
Task:

1. **Setup** (0.5P)
   Import `torch` and `numpy`, then print the installed PyTorch version.

2. **Create tensors** (1P)
   Create a tensor `a` from the list `[1, 2, 3]` and a tensor `b` of ones with shape `(2, 3)`. Print both tensors and their shapes.

3. **Basic operations** (1P)
   Create two float tensors `x = [1.0, 2.0, 3.0]` and `y = [4.0, 5.0, 6.0]`. Compute and print element-wise addition and multiplication.

4. **NumPy ↔ PyTorch conversion** (1.5P)
   Create `np.array([10, 20, 30])`, convert it to a PyTorch tensor, convert it back to NumPy, and print all objects with their Python types.

5. **Reshape** (1.5P)
   Create `t = torch.arange(12)`. Reshape to `(3, 4)`, then reshape back to `(12,)`. Print values and all shapes.

6. **Permute dimensions** (1.5P)
   Create `img = torch.rand(3, 4, 5)` and interpret it as `(C, H, W)`. Permute to `(H, W, C)` and print both shapes. Optionally verify one value before/after permutation.

7. **Squeeze and unsqueeze** (1P)
   Create a tensor `z` with shape `(1, 3, 1, 5)`. Apply `squeeze`, then `unsqueeze(dim=0)`, and print the shape after each step.

Important Note: Submit exactly one ZIP file via Moodle before the deadline. The ZIP file should contain your executable code and your report in PDF format. Make

sure that it runs on different operating systems and use relative paths. Non-trivial sections of your code should be explained with short comments, and variables should have self-explanatory names. The PDF file should contain your written code, all figures, explanations and answers to questions. Make sure that plots have informative axis labels, legends, and captions. Missing plots will result in point reduction.